

SKILL WEEK 7

Task 1:

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t
    pid;
    int status;
    pid =
    fork();
    if (pid < 0) {
        perror("fork failed");    return 1;
    }
    if (pid ==
    0) {

        // Child process    printf("Child process
        started.\n");    printf("Child
        PID: %d\n", getpid());

        sleep(3);

        printf("Child process completed.\n");

        exit(0);

    } else {

        // Parent process    printf("Parent process waiting
        for child...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {    printf("Child exited
        normally.\n");    printf("Child exit status: %d\n",
        WEXITSTATUS(status));
        }

        printf("Parent process completed.\n");
    }
}
```

```
    return 0;
}
teja27@TEJA:~$ nano skill7.1.c
teja27@TEJA:~$ gcc skill7.1.c -o skill7.1
teja27@TEJA:~$ ./skill7.1
Parent process waiting for child...
Child process started.
Child PID: 1363
Child process completed.
Child exited normally.
Child exit status: 0
Parent process completed.
teja27@TEJA:~$ █
```

Task 2:

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

int main() {

    char *path = getenv("PATH");

    if (path == NULL) {        printf("PATH variable
not found.\n");        return 1;
    }

    printf("PATH:\n%s\n\n", path);

    char *copy = strdup(path);    char
*directory = strtok(copy, ":");

    char filename[100];

    printf("Enter executable name to search: ");
    scanf("%99s", filename);

    while (directory != NULL) {

        char fullpath[500];

        snprintf(fullpath, sizeof(fullpath),
            "%s/%s", directory, filename);
```

```

        if (access(fullpath, X_OK) == 0) {
printf("Executable found: %s\n", fullpath);
        free(copy);
return 0;
        }

        directory = strtok(NULL, ".");
    }

printf("Executable not found in PATH.\n");

free(copy);

return 0;
}

```

Output:

```

teja27@TEJA:~$ nano skill7.2.c
teja27@TEJA:~$ gcc skill7.2.c -o skill7.2
teja27@TEJA:~$ ./skill7.2
PATH:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program
Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/wind
ows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/
System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files (x86)/Windows Kits/10/Windows Performance Toolkit:/
mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/tejas/AppData/Local/Programs/Python/Python310/Scri
pts:/mnt/c/Users/tejas/AppData/Local/Programs/Python/Python310:/mnt/c/Users/tejas/AppData/Local/Microsoft/WindowsApps:
/mnt/c/CProg/S10C/bin:/mnt/c/CProg/S10C/bin:/mnt/c/Users/tejas/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users
/tejas/AppData/Roaming/npm:/mnt/c/Program Files/Java/jdk-26/bin:/snap/bin

Enter executable name to search: ls
Executable found: /usr/bin/ls
teja27@TEJA:~$

```